MoLPC: assemble what AlphaFold can predict

Predicting the structure of large protein complexes using AlphaFold and Monte Carlo tree search

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The question: can local predictions determine a large assembly?

2022 study setting
AlphaFold can predict small interacting subcomplexes, while whole complexes with many chains can exceed the tested computational budget.

1. Predict tractable pieces from sequences.
2. Reuse their relative chain orientations to build a large structure.
3. Search among incompatible assembly choices without knowing the native structure.

The hard part is global compatibility, not simply collecting good dimers.

Introduction; Results: Complex assembly. Capabilities refer to the 2022 implementation.

What MoLPC takes as input and returns

Required
Protein sequences and stoichiometry: which proteins, and how many copies of each.

Optional in guided experiments
A native interaction graph indicating which chains contact each other.

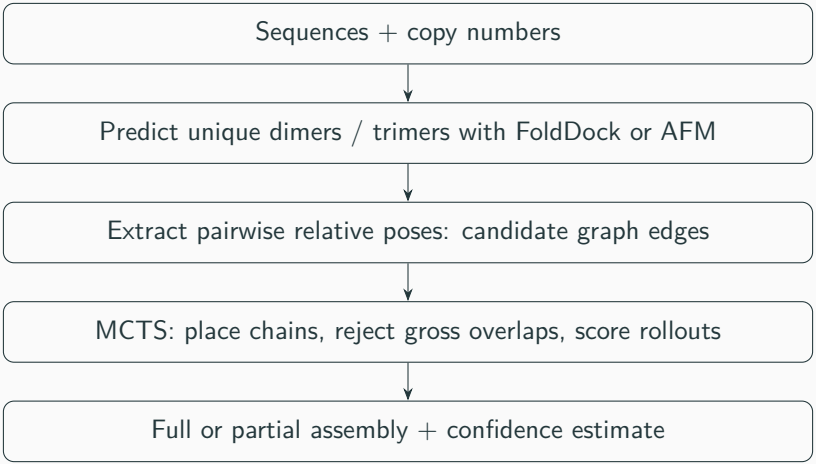
Returned
An assembled coordinate model, possibly incomplete; its confidence score.

Blind all-trimer protocol
Enumerate candidate trimers without supplying native pairwise contacts.

"Blind" does not mean discovering composition or copy number.

Results; Discussion: stoichiometry requirement; Methods: subcomponents.

The pipeline separates prediction from assembly



Original schematic based on Figs. 1–2 and Methods.

Predicting a trimer gives several candidate interfaces

FoldDock in this study

HHblits against Uniclust30; species-matched paired MSA plus a block-diagonal MSA. AlphaFold model_1, 10 recycles, one ensemble.

AFM in this study

Five models, one prediction per model; use the top-ranked structure. Broader database searches and paired/unpaired MSAs.

A predicted trimer *ABC* contributes the poses *AB*, *AC*, and *BC*. The pose *AB* from *ABC* may differ from *AB* from *ABD*.

These are alternative placements to search over; the final structure need not preserve every parent trimer.

Methods: Structural predictions; Path complexity; Results: native trimer assembly.

Running example: superpose the shared chain



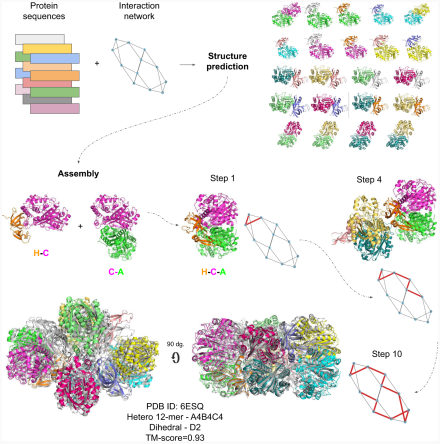
For 6ESQ, start from predicted AC and CH. Fit the coordinates of their shared chain C using SVD superposition; apply that rotation and translation to H.

Continue with HL (align H), then LJ (align L), and so on.

Key assumption: the shared chain provides a compatible rigid alignment anchor. Internal conformational disagreements can accumulate.

Fig. 1 and Methods: Assembly procedure. Diagram shows connectivity, not molecular geometry.

6ESQ: from local poses to a hetero-12-mer



Bryant et al., Fig. 1 (reproduced, CC BY 4.0); full figure retained.

Authors' illustrated guided assembly

Acetoacetyl-CoA thiolase / HMG-CoA synthase complex.

Three protein types, four copies each; dihedral D2 symmetry.

Red graph edges show the chosen assembly path. The final model is overlaid on the native complex in gray.

Figure 1 reports TM-score 0.93 with AFM subcomponents: an existence example, not the benchmark-wide success rate.

The graph alone is insufficient: geometry can invalidate a path

1. A graph edge carries a predicted relative pose, not just an interaction label.
2. Place one new chain through a shared chain already in the assembly.
3. Reject an extension if over 50% of Cα atoms in the shorter of two chains lie within 5 Å of the other chain.

This is a **gross-overlap filter**; it is distinct from the paper's diagnostic atomic clash measure (interchain atom pairs within 1 Å).

In guided experiments, native edges require 10% of residues in the shorter chain to contact the other within 8 Å (Cβ; Cα for Gly).

Methods: Interaction network; Assembly procedure; Clashes.

Why predicting only dimers can make assembly impossible

Consider 1A8R, a homo-decamer: every subunit has the same sequence, but its native neighbors occupy more than one relative orientation.

A single unique sequence pair is not a complete pose library
A dimer prediction can repeatedly recover one interface mode. Reusing it may produce a small closed unit or overlapping chains rather than the full complex.

Predicting trimers can reveal additional compatible interface modes. Extracting their constituent pairs expands the available pose library.

Evidence: native trimers increase completion from 15 to 58 of 175 with FoldDock.

Supplementary Fig. 1c; Results: Limited conformational sampling; Fig. 3a.

MCTS searches partial assemblies, not amino-acid sequences

Tree node
A particular partial assembly / placement history.

Action
Add an unplaced chain using an available predicted pose.

Terminal state
All required chains placed, or no legal extension.

One search iteration
Selection → expansion → simulation → backpropagation.

The published search stops at completion or when no non-overlapping extension remains. It seeks the largest possible assembly; it offers no global-optimum guarantee.

The search statistics belong to this target; backpropagation updates node scores, not AlphaFold network weights.

Figs. 2, 8; Methods: MCTS and pseudocode.

Selection balances good previous outcomes and exploration

For child i, the paper uses

$$\mathrm{UCB}_i = V_i + 2\sqrt{\frac{\ln N}{n_i}}.$$

V_i: mean complex score from continuations below child i.

N: visits to its parent; n_i: visits to child i.

Exploitation
Large V_i favors branches whose rollouts have scored well.

Exploration
Small n_i increases the bonus, encouraging less explored options.

Start from a randomly selected chain; traverse expanded nodes using the best-child rule.

Methods Eq. 5, Fig. 8. Definitions follow the paper.

Expansion and simulation test an early placement

Expansion: create candidate extensions through connecting edges, then randomly extend newly added nodes.

Simulation / rollout: continue adding chains randomly, applying predicted rigid transforms, until complete or blocked by overlap / unavailable edges.

Running example: two possible placements of H
From AC, an H pose supported by ACH may permit L and then J. Another candidate H pose may occupy space needed for L. Rollouts expose that downstream consequence.

This branching example is illustrative; the $AC \rightarrow ACH \rightarrow ACHL$ chain sequence comes from Fig. 1.

Methods: Expansion and Simulation; Fig. 1. Alternative-H scenario is a teaching example.

Backpropagation assigns credit to the assembly history

Score a rollout's resulting structure, then update the statistics of its ancestor nodes.

$$AC \longrightarrow ACH \longrightarrow ACHL \longrightarrow \cdots$$

the rollout score informs the earlier decisions

Repeated rollouts provide support for early choices that tend to permit high-scoring continuations.

A poor rollout does not prove an interface is wrong. The result depends on pose availability, random exploration, score quality, and search time.

Figs. 2, 8; Methods: Backpropagation and pseudocode. Last statement is interpretation.

The actual search reward favors confident, extensive interfaces

For each chain a, collect its interface with *all* other chains:

$$s_a = \overline{\mathrm{pLDDT}}_{\mathrm{interface},a} \log_{10}(C_a), \qquad S = \sum_a s_a.$$

C_a: interchain residue contacts within 8 Å using Cβ (Cα for Gly).

If A contacts both B and C, score its combined interface rather than treating each edge as an independent final objective.

Search optimizes S: higher-confidence and larger interfaces raise the score. Native coordinates and TM-score are unavailable to the search.

Completion can increase reward by adding interfaces; it is not an accuracy guarantee.

Methods: Scoring, Eq. 6. Notation here makes the stated per-chain summation explicit.

mpDockQ confidence and measured accuracy are different objects

Confidence calibration

For interface summary $x = \overline{\mathrm{pLDDT}} \log_{10}(\text{contacts})$, fit

$$\mathrm{mpDockQ}(x) = \frac{0.728}{1 + \exp[-0.098(x - 309.375)]} + 0.262.$$

The fit targets TM-score using complete assembled complexes from this benchmark; this is not an independent calibration test.

Accuracy evaluation

MM-align compares predicted and experimental assemblies to produce TM-score; the paper emphasizes TM-score ≥ 0.8 for close whole-complex agreement.

The name does not make mpDockQ a measured DockQ or a success probability. The paper also calls the raw search score "mpDockQ"; distinguish raw reward from sigmoid calibration.

Methods: Scoring; Sigmoidal fit, Eq. 7; MMalign; Fig. 3b–c.

The large-complex benchmark has a restricted scope

175 nonredundant experimental complexes with 10–30 chains

PDB query in January 2022; X-ray or EM, resolution ≤ 3 Å; no nucleic acids; all chains from one organism.

Sequence clustering at 20% identity removes redundant complex compositions; remove chains shorter than 50 residues, incorrect stoichiometry and discontinuous chains.

Most targets have 10–12 chains.
114 homomers, 61 heteromers.

Dihedral/cyclic symmetries dominate.
Not a random sample of cellular assemblies.

The 20% clustering is dataset nonredundancy; it is not the smaller benchmark's explicit AFM-training-set filter.

Methods: Non-redundant 10–30-chain complexes; Fig. 6; Fig. 4c.

Completion rises with a richer pose library; accuracy need not

	FoldDock		AFM	
Subcomponents	Complete	Median TM	Complete	Median TM
Native dimers	15	0.77	15	0.83
Native trimers	58	0.80	55	0.74
All trimers	91	0.51	74	0.61

All completion counts have denominator **175**. Medians condition on completion.

The blind FoldDock protocol completes **52%**, but only **30/175** are both complete and TM-score ≥ 0.8.

Main-text native-trimer median is shown; Supplementary Table 1 instead reports 0.88 for FoldDock.

Results; Fig. 3a–d; Supplementary Table 1. Main-text all-trimer median 0.51.

A fair comparison must say which targets are being averaged

1. **All targets:** 175 large complexes; failures and incomplete assemblies matter for practical yield.
2. **Complete within each method:** 91 FoldDock all-trimer models versus 74 AFM; their medians compare different target subsets.
3. **Matched complete subsets:** FoldDock native/all-trimer comparison on 53 shared targets gives medians 0.80/0.76.

Figure 3a pools 108 targets completed by at least one approach and inserts zeros for missing methods. Its plotted averages are not averages over the entire 175-target benchmark.

"More complete assemblies" and "more accurate completed assemblies" answer different questions.

Fig. 3a caption; Results: Complex assembly using trimeric interactions.

When does assembly work? Quality of the pieces is decisive

Native-trimer FoldDock analysis

Across 58 complete assemblies, mean subcomponent TM-score and assembled-complex TM-score have Spearman $R = 0.80$.

Accurately predicted pieces are necessary for reliable global placement; MCTS cannot create a missing relative pose.

New interfaces can be worse than directly predicted ones

For 55 completed complexes with assembly-generated interfaces, mean DockQ is 0.34 for directly predicted interfaces versus 0.24 for newly assembled interfaces.

Interpretation: local rigid placements can compound errors even when each individual contact looks plausible.

Fig. 4f; Results: Aspects affecting assembly; Supplementary Fig. 5. Correlation is not causation.

Symmetry helps, but a completed symmetric model can be wrong

All-trimer FoldDock: symmetry	Complete	Median TM
Dihedral	36	0.75
Cyclic	31	0.54
Tetrahedral	6	0.93
Asymmetric	15	0.38
Helical	2	0.60
Octahedral	1	0.22

Counts sum to 91 complete assemblies; small groups support little generalization.

Example: 5TRM is annotated octahedral but assembled into a dihedral configuration (TM-score 0.22). Stoichiometry and contacts alone need not identify the native global arrangement.

Supplementary Fig. 3; Fig. 3d and Results. These counts concern the blind all-trimer protocol.

Smaller complexes test the value of assembly against end-to-end AFM

278 targets with 4–9 chains

50 targets per size from 4–8 chains, plus 28 nonamers; released after the AFM training cutoff and filtered against training complexes at 30% sequence identity.

FoldDock all-trimer assembly fails to complete 62; AFM end-to-end fails on memory for 67.

The authors conclude that AFM end-to-end is more accurate when it can be run. The assembly strategy addresses computational reach rather than a universal accuracy improvement.

Figure 5 represents missing outputs as zero. Hardware failures therefore contribute to the plotted performance.

Fig. 5; Methods: 4–9-chain set and prediction limitations; Discussion. See README for numeric-label caveat.

Compute is shifted to many smaller predictions

Combinatorics

n chain copies yield $\binom{n}{3}$ trimers before sequence redundancy reduction.

At $n = 30$: 4,060 trimers, each offering three pairwise poses.

The global placement search still grows rapidly.

Reported 2022 cost

FoldDock: roughly 1–2 h per subcomponent; AFM: 3–4 h.

All-trimer assembly averages 3.73 CPU h per target; six targets exceed a 72 h search limit.

Large trimers can still exceed GPU memory/time budgets.

Prediction dominates average cost; "CPU-only assembly" does not make the whole pipeline inexpensive.

Methods: Complexity; Computational time; Structural prediction limitations; MCTS. Hardware-specific estimates.

What the baseline comparisons establish

HADDOCK

Given native chain structures, 77 complete models; median TM-score 0.29, none ≥ 0.5. This used random-surface / center-of-mass restraints rather than target-specific experimental restraints.

Multi-LZerD

No completed target within the authors' allocation: even a 10-chain test exceeded 72 h. This is a resource-limit observation, not proof of intrinsic inability.

Takeaway

MoLPC demonstrated a useful sequence-based route in this benchmark; it does not replace experimentally guided integrative modeling.

Results; Methods: Comparison with other methods; Supplementary Fig. 2.

What remains unresolved?

1. **Input uncertainty:** composition and stoichiometry are required; cellular partners and copy numbers may be unknown.
2. **Geometry:** rigid superposition and a permissive overlap filter do not optimize all global interfaces or distinguish alternative biological states.
3. **Confidence:** mpDockQ is fitted on assembled benchmark models; independent calibration and prospective validation remain important.

Discussion prompt: Which additional information would be most valuable: better local pose diversity, global symmetry/shape restraints, or a better reward for partial assemblies?

The result is a proof of feasibility: **predict local pieces, then search globally**, with correct full assembly still achieved for a minority of targets.

Discussion; Methods; Figs. 3–4. Calibration/generalization points are journal-club interpretation.